



جامعة الأمير ساطم بن عبد العزيز
PRINCE SATTAM BIN ABDULAZIZ UNIVERSITY

Prince Sattam bin Abdulaziz University
College of Computer Engineering and Sciences
Software Engineering Project

FIT World Systems Project

Group members:

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1. Group Project Proposal

A. Introduction

Healthy living is built on a foundation of good nutrition, regular exercise, sufficient sleep, and staying well-hydrated. Living a healthy lifestyle keeps you fit, energetic, and lowers the risk of various diseases. Adopting a healthy way of life allows you to enjoy more aspects of your life and reduces the chance of serious illness or early death. Health is about more than just avoiding sickness—it encompasses physical, mental, and social well-being.

By choosing to live a healthy lifestyle, you set a positive example for others around you, especially your family and children. You also create a healthier environment for them to grow up in. Supporting them in following healthier habits contributes to their well-being and quality of life, both now and in the future.

Our application is dedicated to providing the information and services that people need to stay active or begin a healthier lifestyle journey.

B. Statement of Problem

In today's fast-paced society, maintaining a healthy lifestyle is a challenge faced by many individuals, often due to boredom or a lack of inspiration. In addition, many people lack the knowledge necessary to keep track of their calories or prepare wholesome meals. Not to mention those who intend to consume the ideal amount of water but fail to do so.

After giving these issues some thought, we made the decision to create an app aims to empower individuals to achieve their health goals more effectively.

C. Background Survey

We noticed that most applications focus only on the exercise aspect, providing suitable workouts and required hours, but they do not address the importance of the dietary aspect, which requires attention to calculating caloric intake and adhering to meals that meet a person's daily needs. This also includes calculating the appropriate amount of water, an important factor for achieving the best results. We can also offer methods for preparing healthy meals to diversify options, so they don't become routine. Therefore, we decided to create an application that fulfills these requirements.

D. Proposed Approach

In response to the challenges individuals often encounter in maintaining healthy habits, we propose an app that delivers motivational notifications to remind users to drink water, eat meals, and walk at designated times. By tailoring these reminders to each user's lifestyle, age, and weight based on information collected through a series of onboarding questions the app aims to offer a personalized experience that effectively promotes healthy habits.

E. Work Plan

Project Parts	Task	Start	End	Status
Part 1	Choose topic	27/10	1/11	Done
Part 2	Introduction	3/11	7/11	
	Statement of Problem	3/11	7/11	
	Background Survey	3/11	7/11	
	Proposed Approach	3/11	7/11	
	Work Plan	3/11	7/11	
Part 3	Functional requirements	16/11	16/11	
	non- Functional requirements	16/11	16/11	
Part 4	Diagram	16/11	16/11	
	Senarios	16/11	16/11	
Part 5	Creating Sequence Diagrams	16/11	16/11	
Part 6	Creating a Class Diagram	16/11	16/11	
Part 7	User Interface Design	20/11	20/11	
Part 8	Demo	20/11	20/11	

2. Group Project requirements

A. functional requirements

ID	Priority	Description
R1	High	The user shall be able to register a new account with personal details (e.g., username, email, password) to personalize the experience and securely store their fitness data.
R2	High	The user shall be able to log in and log out of their account to access their personal fitness information and ensure data privacy.
R3	High	The user shall be able to set fitness goals (e.g., daily steps, calories burned) and track their progress towards these goals over time to maintain motivation.
R4	Low	The user shall be able to add and manage friends to share fitness achievements and motivate each other, creating a social and supportive environment.
R5	Mid	The user shall be able to add custom exercises that are not predefined in the app, allowing them to tailor their fitness routines according to their preferences.

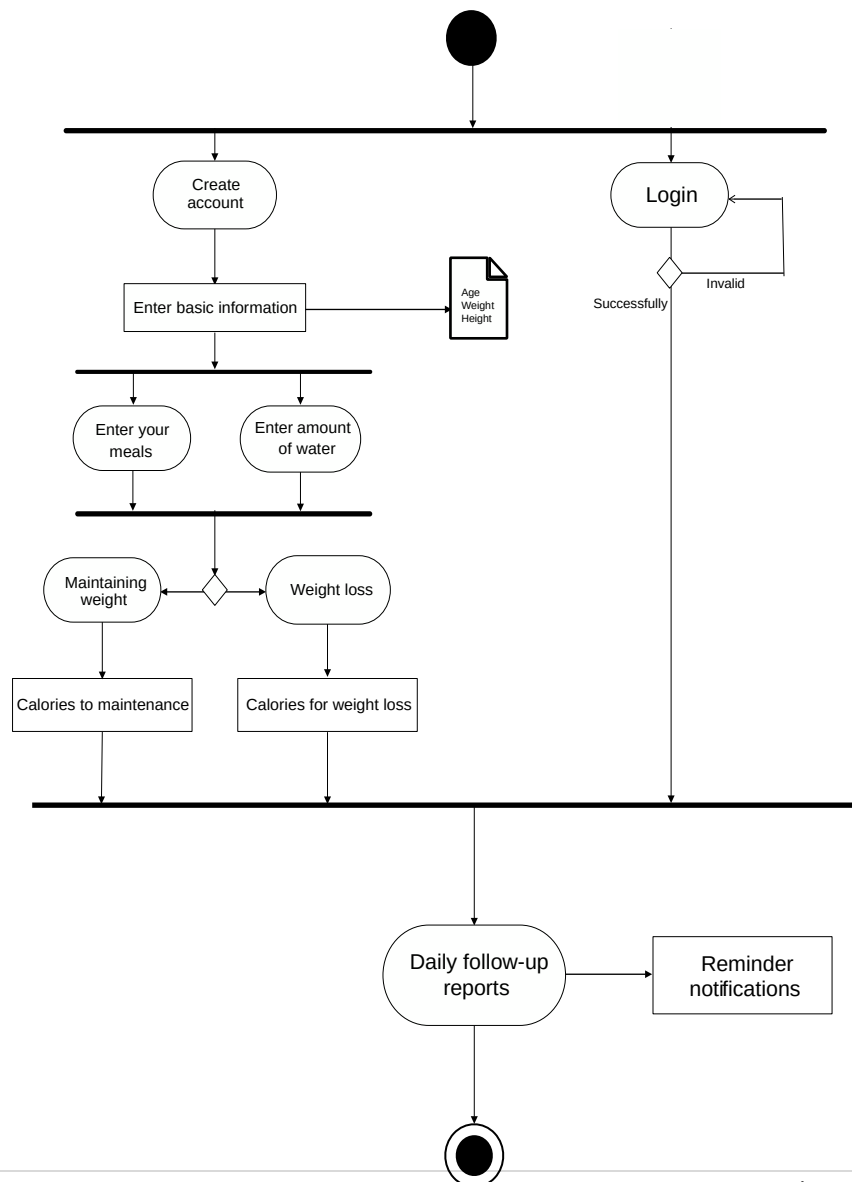
ID	Priority	Description
R6	Mid	The system shall send reminder notifications to users to encourage them to meet their daily fitness goals (e.g., drink water, complete exercises)
R7	Mid	The system shall allow users to communicate in real-time with professional trainers for guidance and motivation, enhancing user experience and support.
R8	Mid	The system shall allow users to communicate in real-time with professional trainers for guidance and motivation, enhancing user experience and support.

B. non-functional requirements

ID	Priority	Description
R9	High	The system should be able to handle up to 10,000 concurrent users without performance degradation to support a high number of active users.
R10	Mid	Activity tracking data from connected wearable devices should sync with the app within 5 seconds to ensure up-to-date information for users.
R11	High	All user health and fitness data must be securely stored using encryption to protect sensitive information from unauthorized access.
R12	High	The system shall require two-factor authentication (2FA) for user account creation and login to enhance account security.

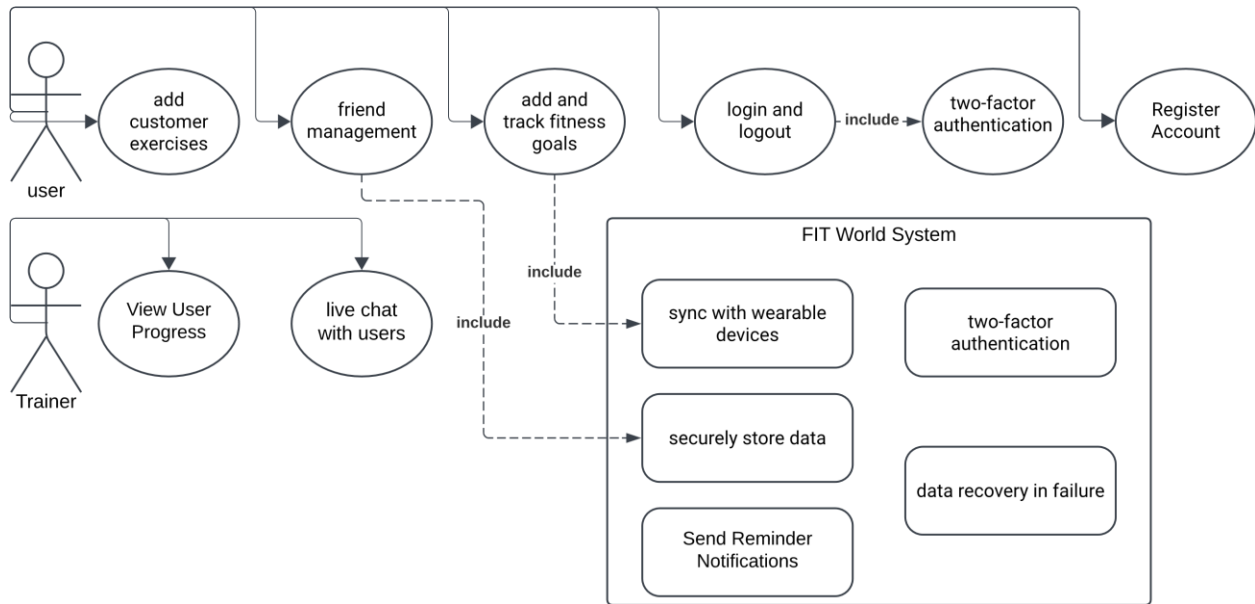
R13	High	In the event of a system failure, all user data must be restored within 30 minutes to minimize data loss and ensure system reliability.
R14	Mid	The system shall perform regular automated data backups every 6 hours to maintain data safety.
R15	High	The user interface (UI) should be easy to navigate, with clear instructions for setting goals and tracking activities, providing a user-friendly experience.
R16	Mid	The application shall provide onboarding tutorials for new users to help them quickly understand and use the main features effectively.

3. Project Activity diagram (Process Model)



4. Project Use Case Modelling

A. Diagram:



B. Senarios:

1. Table1:

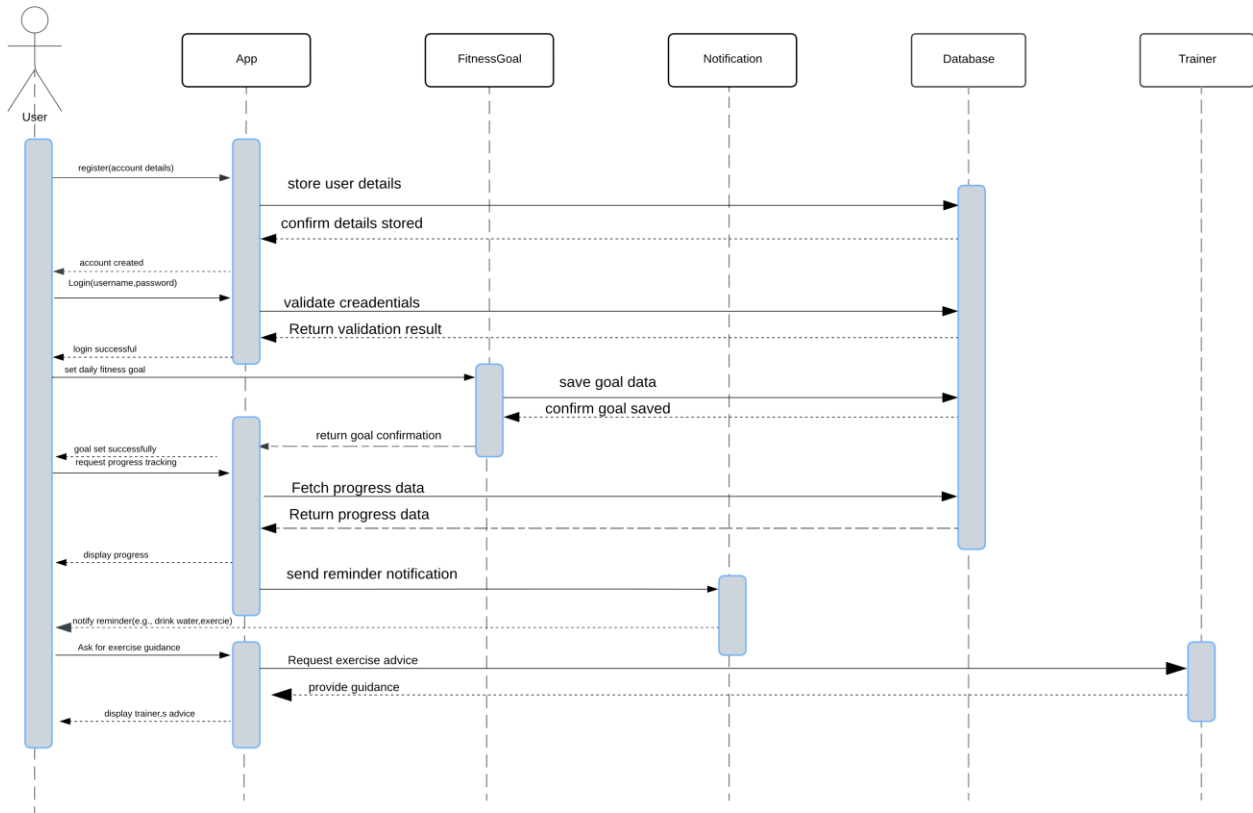
Use Case Name	Register account
Actor	User
Difficulty	Easy
Pre-condition	The user has downloaded the app and wants to create an account.
Post-condition	The user account is created with personal details stored securely.
Processes	The user enters details such as username, email, and password. The system validates the details and stores them in the database.

2. Table2:

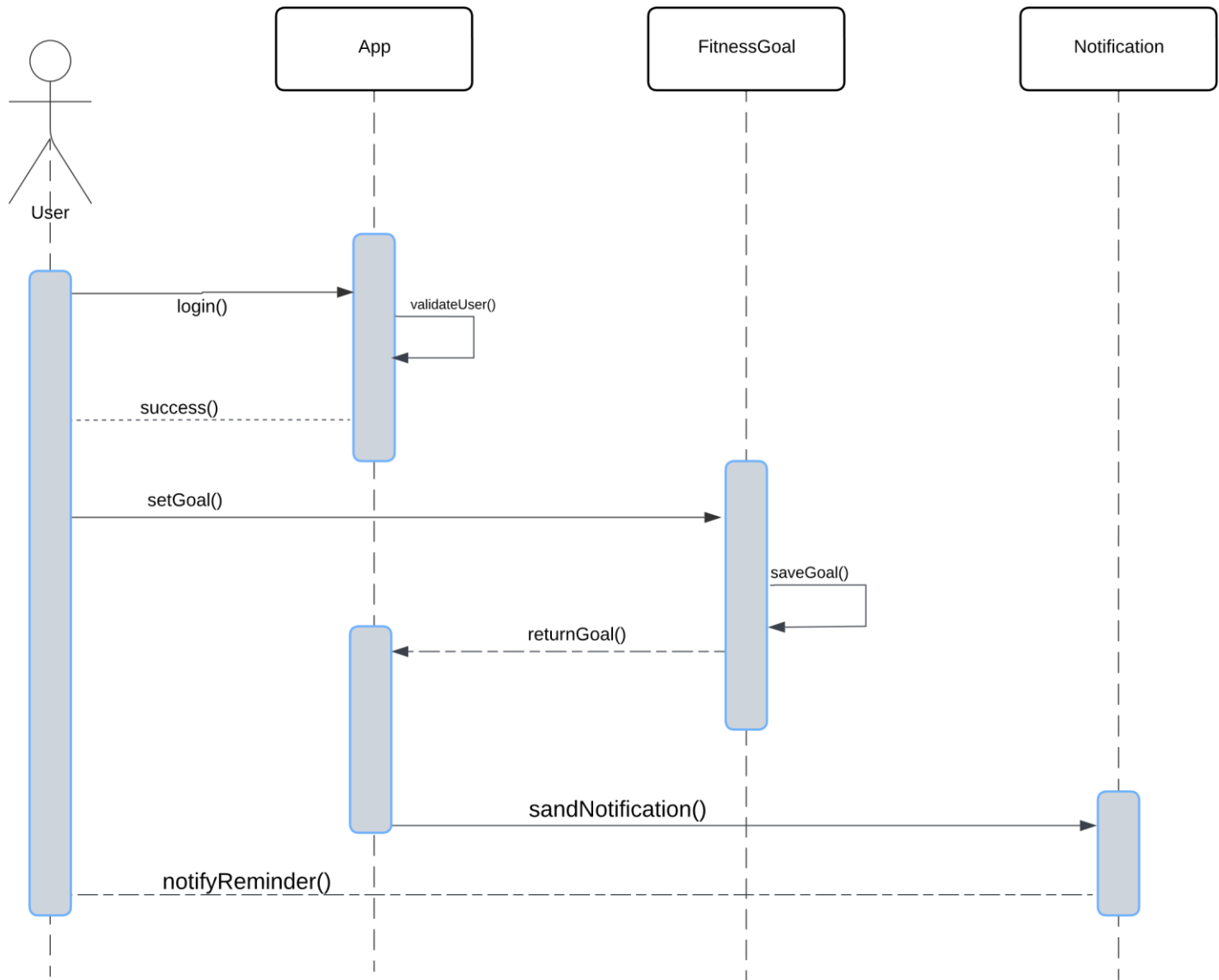
Use Case Name	Send reminder notifications
Actor	System
Difficulty	Intermediate
Pre-condition	The user has set daily goals.
Post-condition	The user receives notifications to stay on track with their fitness goals.
Processes	The system sends push notifications based on the user's preferences (e.g., reminders to exercise or drink water).

5. Creating Sequence Diagrams

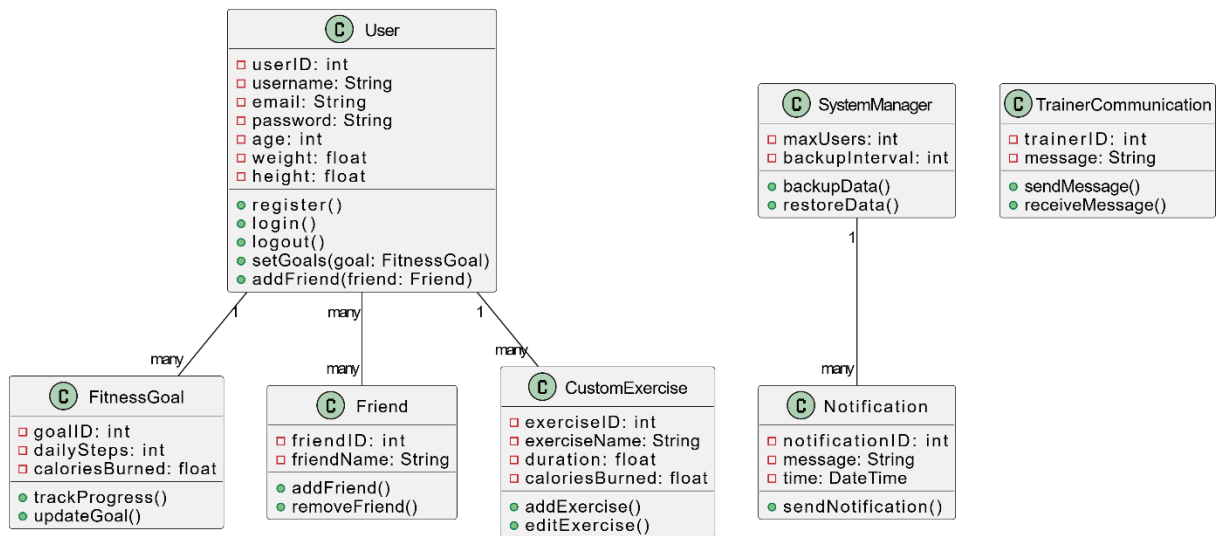
A. Diagram1:



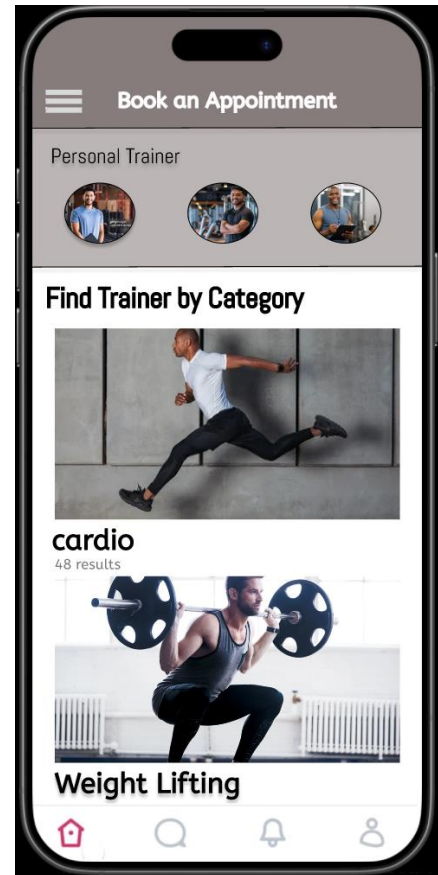
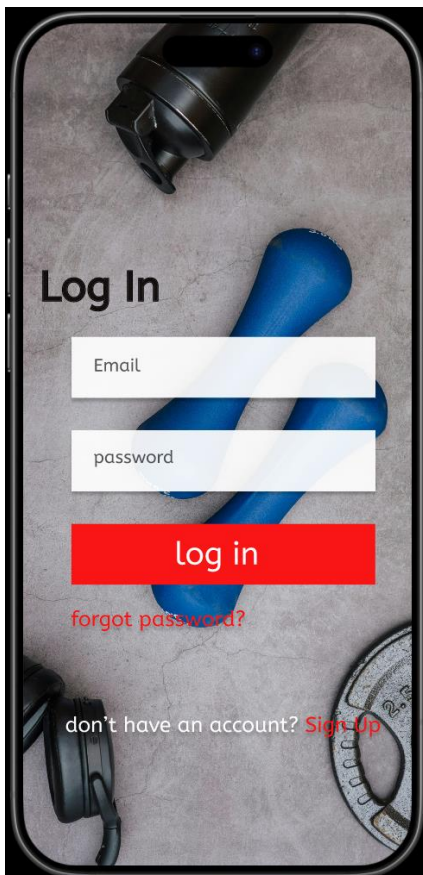
B. Diagram2:

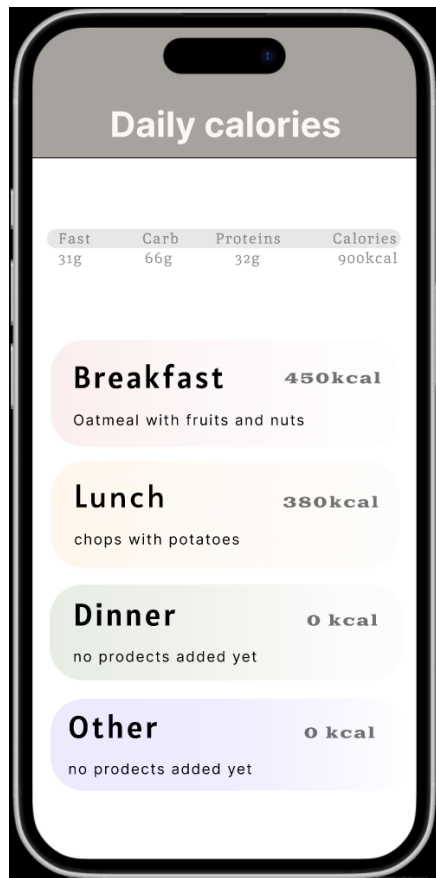


6. Creating a Class Diagram



7. User Interface Design





Marking Schema

Group Number: 3

Total Mark: /20

<u>Parts</u>				Total	Final
1: Feasibility Study &Project Proposal:				2	
Introduction	0.5	solutions	0.5		
Problems	0.5	Work Plan	0.5		
similar ideas		Overall			
2: Group Project requirements:				2	
Write the FR perfectly (8) ?	.5				
Write the NFR perfectly (8)?	.5				
3: Project Activity diagram (Process Model):				2	
The flow of most processes	1				
Start point / End point / Decision paths	1				
4: Project Use Case Modelling:				2	
Actors	.5				
Use cases &its related use cases	.5				
Table 1	0.5				
Table 2	0.5				
5: Creating Sequence Diagrams				2	
Objects	1				
Messages	1				
Overall					
6: Creating a Class Diagram:				2	
Classes+ attributes+ operations	1				
associations	1				
overall					
7: User Interface Design:				2	
8: Report contents are accurately prepared as per the technical specification of a required report:				1	
9: Presentation				5	